Phase 13: Containerization & Kubernetes Deployment

Course Code: 20CYS401 — Secure Software Engineering

Project: Online Book Review and Recommendation Platform

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1. Hardened Dockerfile Architecture

The container runtime uses a multi-stage build pattern (`docker/Dockerfile`):

- **Base Image**: `python:3.11-slim`

- **Builder Stage**: Compiles dependencies in isolated cache layer.

- **Runtime Stage**:

- Unprivileged non-root user `appuser` (`UID 10001:10001`).

- Read-write restricted instance folder for SQLite.

- Active healthcheck probe inspecting `http://localhost:5000/health`.

- Zero hardcoded secrets or environment credentials in image layers.

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2. Local Docker Compose Verification

- **Compose Configuration**: `docker-compose.yml`

- **Port Mapping**: `5050:5000` (mapped to host port 5050 to prevent macOS AirPlay port 5000 conflicts).

- **Health Verification**:

```
curl http://localhost:5050/health
```

Returns:

```
{"database":"connected","service":"secure-book-review-platform","status":"healthy"}
```

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3. Kubernetes Manifests

Located in `k8s/`:

- `namespace.yaml`: Isolated namespace `bookplatform`.

- `deployment.yaml`: Enforces `runAsNonRoot: true`, `readOnlyRootFilesystem: false`, CPU/Memory resource bounds, and liveness/readiness HTTP probes on `/health`.

- `service.yaml`: ClusterIP internal service exposure.